版本号	修订日期	修订记录
V1.0	2025-12-19	初始版本。

SEEKFREE

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在此文件中，仅以版本号前两位(发行版)，划分版本之间的区别。

Title: STC32G144K246主板			Author: 硫酸铜
Subtitle: Revision			Revision: V1.0.3.0
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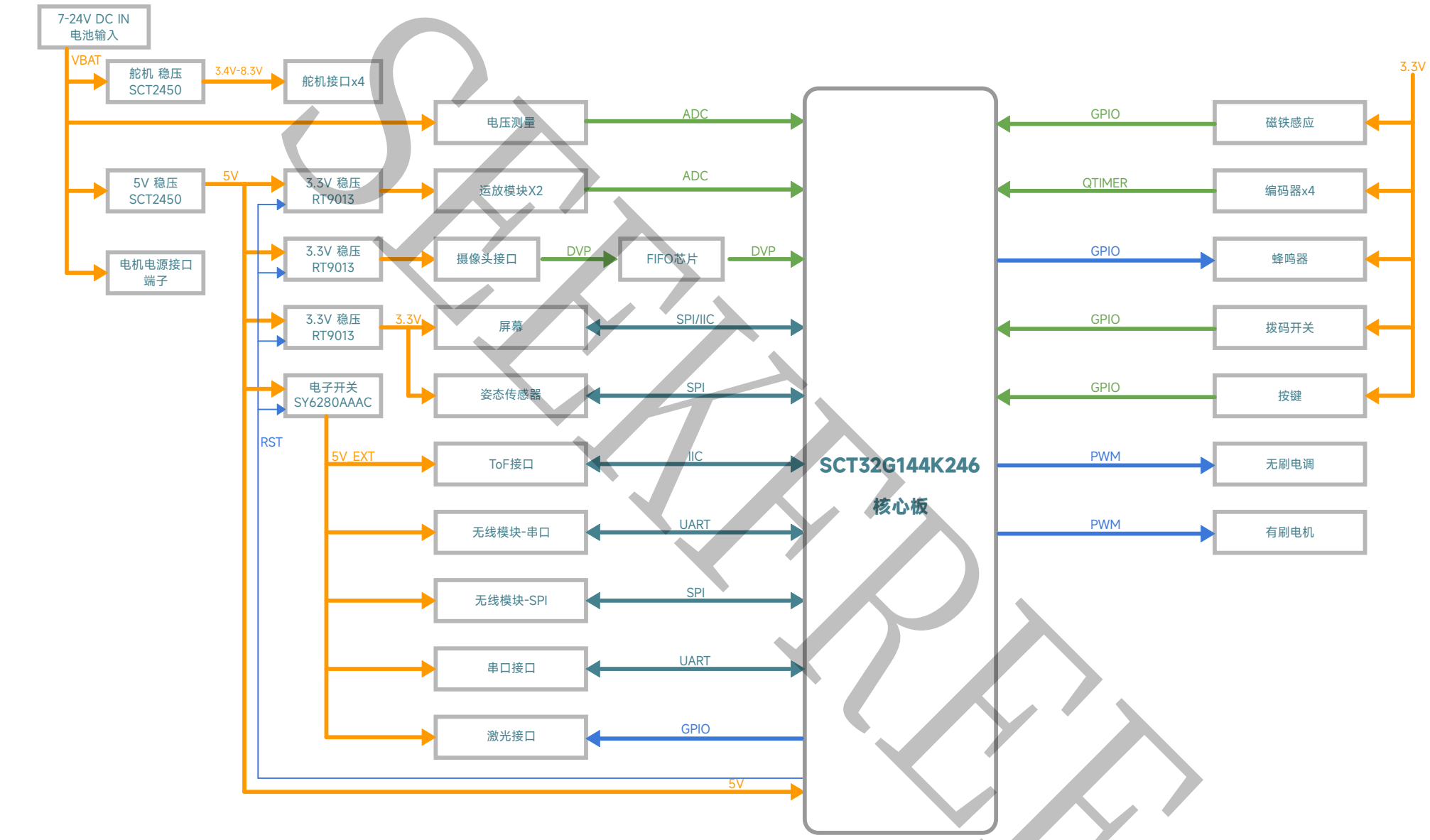

图例：

双向信号：UARTx　输出信号：UARTx　输入信号：UARTx　电源：5V

红色字体：需要注意/未确定条目

Title: STC32G144K246主板			Author: 硫酸铜
Subtitle: BlockDiagram			Revision: V1.0.3.0
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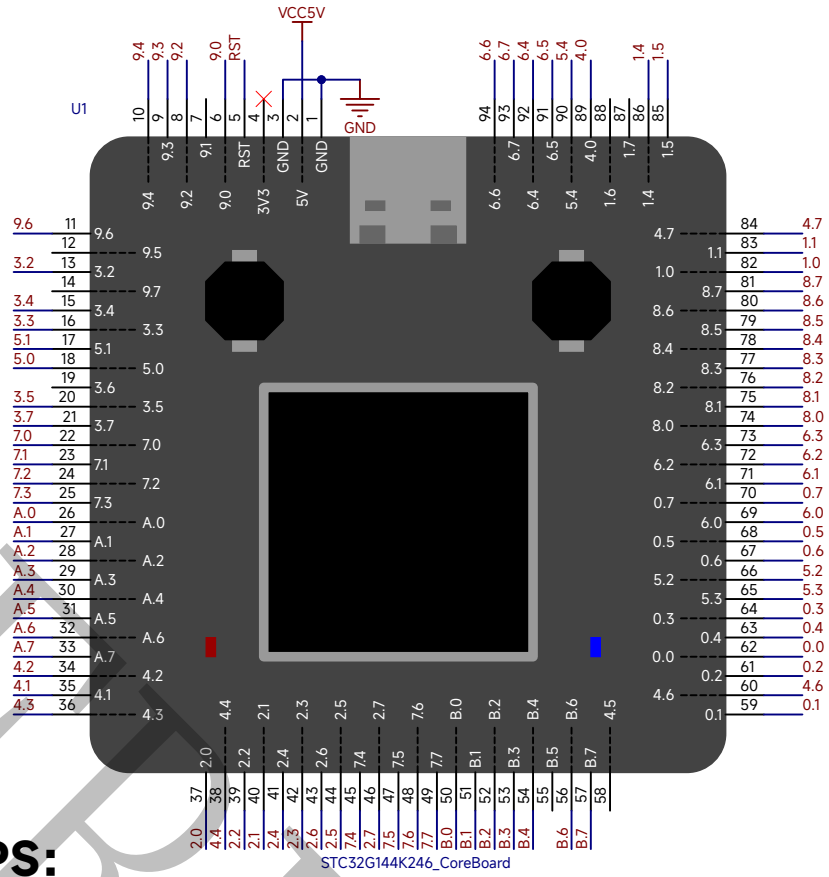
外设	功能	对应引脚
摄像头	VSY	3.3
	SDA	B.7
	SCL	B.6
	WEN	4.1
	WRST	7.3
	RCK	3.7
	RRST	4.3
	OE	7.2
	D0-D7	2.0-2.7
编码器	A	4.0
	B	4.2
	A	6.0
	B	6.2
	LSB	8.0
	DIR	4.4
	LSB	9.0
	DIR	4.6
运放	ADC	0.0
	ADC	0.1
	ADC	0.2
	ADC	0.3
	ADC	0.6
	ADC	0.7
	ADC	1.1
	ADC	1.4
电池电压	ADC	1.0
屏幕接口	SCK	8.3
	MOSI	8.1
	RST	7.0
	DC	7.1
	CS	3.5
	MISO	8.2
ToF接口	SCL	6.1
	SDA	6.3
	XS	6.6
	INT	8.4
姿态传感器	SCK	8.7
	MOSI	8.5
	MISO	8.6
	CS	3.4

外设	功能	对应引脚
串口	Tx	0.5
	Rx	0.4
无线模块串口	Tx	9.3
	Rx	9.2
	RTS	9.4
	CMD	5.4
无线模块SPI	MISO	8.2
	SCK	8.3
	MOSI	8.1
	CS	6.4
	INT	1.5
	RST	4.7
有刷电机	PWM	7.4
	PWM	7.5
	PWM	7.6
	PWM	7.7
	PWM	5.0
	PWM	5.1
	PWM	5.3
	PWM	5.2
无刷电调	PWM	A.1
	PWM	A.3
	PWM	A.5
	PWM	A.7
舵机	PWM	A.0
	PWM	A.2
	PWM	A.4
	PWM	A.6
拨码开关	IO	B.0
	IO	B.1
按键	IO	B.2
	IO	B.3
	IO	B.4
	IO	3.2
蜂鸣器	IO	6.5
激光	IO	6.7
磁铁感应	IO	9.6

引脚复用

引脚表



TIPS:

1.主板上的所有外设（除舵机外），均使用RST引脚来控制上电时序，

2.主板上的屏幕接口和无线模块SPI引脚复用，两者不可同时使用

3.激光接口为IO口控制电源，需要使用时请将P6.7提供高电平来开启
不使用单片机时将会无法使用，此时可以通过杜邦线将P6.7与3V3连接来测试

4.主板摄像头接口的2.0*-2.7*表示摄像头数据经过FIFO芯片再读取至单片机IO口，
此处带*号仅提示对应使用的IO口读取数据，不代表直接与核心板相连接

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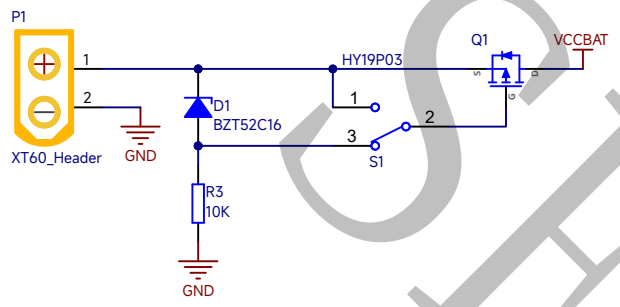
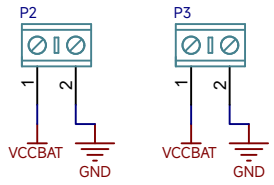
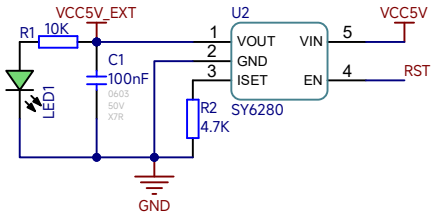
电源接口/开关



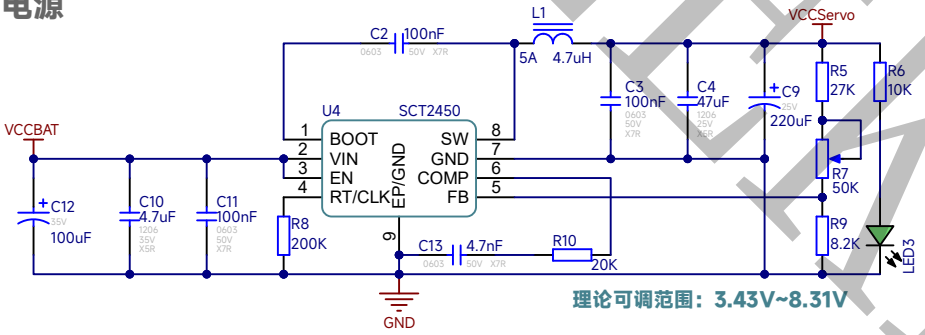
电机电源接口



5V外设电源控制

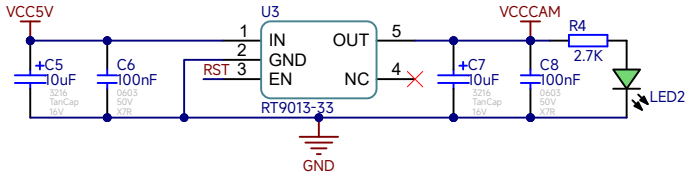


舵机电源

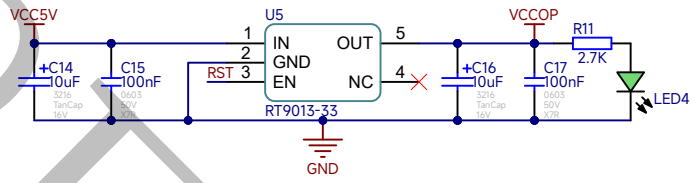


理论可调范围：3.43V~8.31V

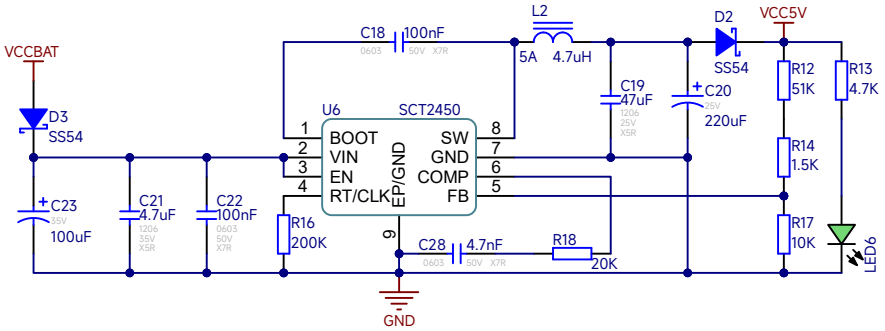
摄像头电源



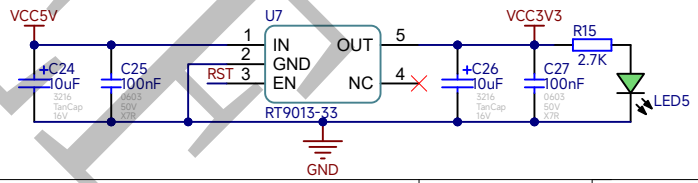
运放电源



5V电源



3V3电源



Title: STC32G144K246主板		**Author:** 硫酸铜	逐飞科技 SEEKFREE
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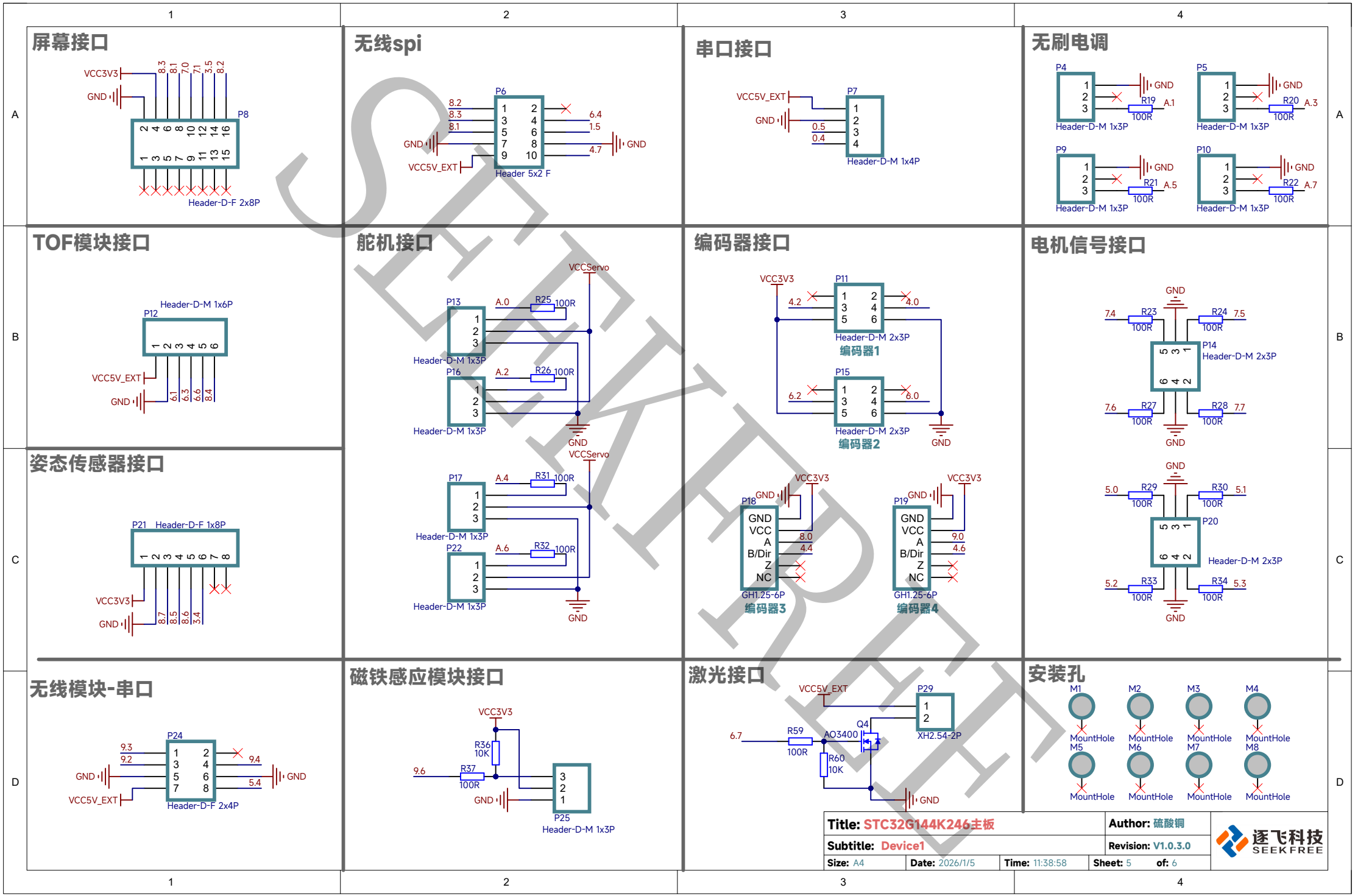
屏幕接口
VCC3V3
GND
P8
Header-D-F 2x8P
无线spi
P6
Header 5x2 F
VCC5V_EXT
GND
串口接口
P7
Header-D-M 1x4P
无刷电调
P4
P5
P9
P10
Header-D-M 1x3P
R19
R20
R21
R22
100R
TOF模块接口
P12
Header-D-M 1x6P
舵机接口
VCCServo
P13
P16
P17
P22
R25
R26
R31
R32
编码器接口
P11
P15
Header-D-M 2x3P
编码器1
编码器2
P18
P19
GH1.25-6P
编码器3
编码器4
电机信号接口
P14
P20
姿态传感器接口
P21
Header-D-F 1x8P
无线模块-串口
P24
Header-D-F 2x4P
磁铁感应模块接口
R36
R37
P25
激光接口
P29
XH2.54-2P
Q4
AO3400
R59
R60
安装孔
M1
M2
M3
M4
M5
M6
M7
M8
MountHole
Title: STC32G144K246主板
Author: 硫酸铜
Subtitle: Device1
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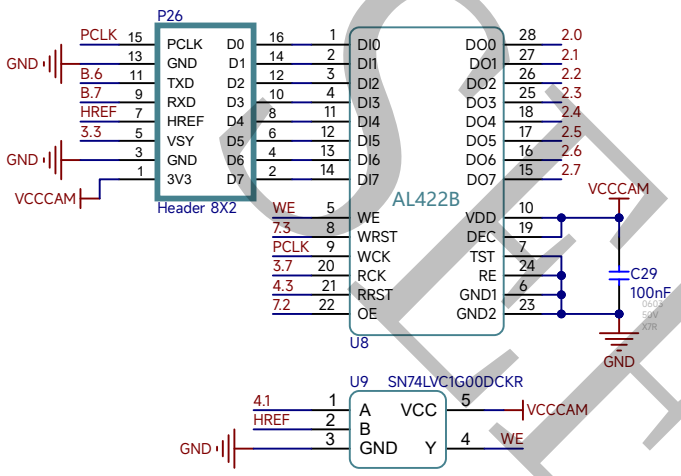
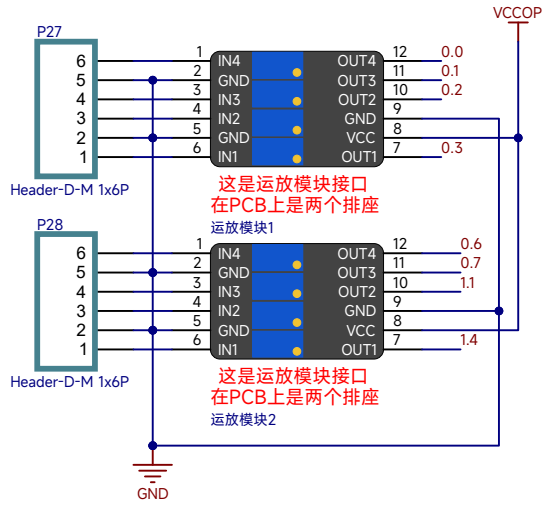
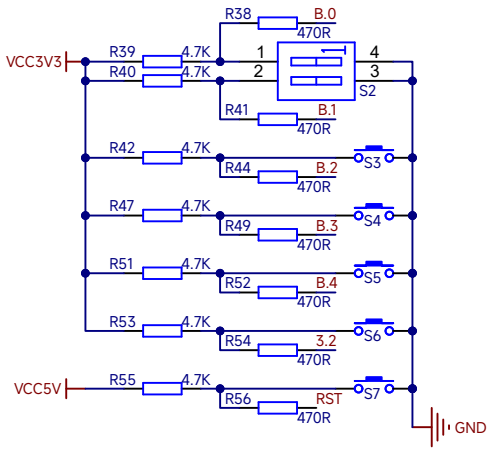
摄像头接口



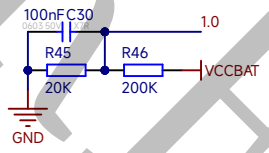
电磁运放模块



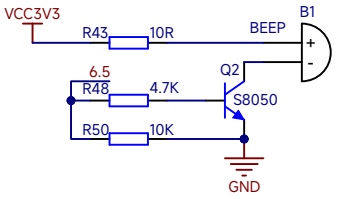
拨码开关/按键



电压测量



蜂鸣器



Title: STC32G144K246主板		**Author:** 硫酸铜	逐飞科技 SEEKFREE
Subtitle: Device2		**Revision:** V1.0.3.0	
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